

Utilization of environmental and epidemiological indicators in the study of malaria dynamics

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ABSTRACT

This paper aims to analyze the behavior of malaria transmission in the Amazon region based on climatic and environmental changes, such as temperature, precipitation, deforestation and forest fires, through proposed modifications to the SIR and SEI models, in order to contribute to the study of applications of external effects on the evolution of the disease. The Trajetorias Project, developed by the Synthesis Center on Biodiversity and Ecosystem Services (SinBiose/CNPq) was used as an initial reference for the study.

This work employs a modified SEIRS/SEI methodology, based on the SIR/SEI model developed by Parham and Michael (2010) which takes into account rainfall and temperature, with further modifications.

1. Introduction

Malaria remains one of the most significant vector-borne diseases worldwide, caused by protozoa of the genus *Plasmodium* and transmitted by female mosquitoes of the genus *Anopheles*. Although present in multiple regions, the burden of malaria in the Americas is highly concentrated in the Amazon basin, which spans nine countries and accounts for a substantial proportion of reported cases [1]. The persistence of malaria in this region is closely linked to ecological, climatic, and socioeconomic factors that influence both vector dynamics and human exposure.

Environmental changes play a central role in shaping malaria transmission. In the Amazon, processes such as deforestation, agricultural expansion, and infrastructure development alter local ecosystems, affecting vector habitats and increasing human-vector contact. These changes often promote the proliferation of mosquito breeding sites and concentrate human populations in high-risk areas, particularly among non-immune migrants [4, 5]. In addition to land-use change, climatic variables such as temperature and rainfall, as well as disturbances such as forest fires and mining activities, further influence vector abundance, pathogen development, and transmission intensity.

Despite the recognized importance of environmental drivers, integrating these factors into mathematical models of malaria transmission remains challenging. In particular, distinguishing between their effects on mosquito biology and their impact on human-vector interaction is essential for developing models that are both biologically consistent and identifiable from available data.

In this work, we develop and analyze a modified SIR-SEI

model for malaria transmission in the Amazon, incorporating environmental variables such as temperature, rainfall, deforestation, and fire activity. Climatic factors are modeled as drivers of mosquito life-history parameters, while land-use variables are introduced as modifiers of transmission intensity. Using epidemiological and environmental data from the Trajetorias Project [6], we investigate how these factors influence disease dynamics and assess their relative contributions through analytical and numerical methods.

2. Model formulation

2.1. Base SIR-SEI model

Based on previous work developed to model the transmission of malaria based on precipitation and temperature dynamics [7], we initially focused on two different sets of compartments of hosts and vectors: susceptible, infected and recovered hosts (S_H , I_H , R_H respectively), and susceptible, exposed and infected vectors (S_M , E_M , I_M respectively).

The equations that describe the transmission are initially given as:

$$\frac{dS_H}{dt} = -ab_2 \left(\frac{I_M}{N} \right) S_H \quad (1)$$

$$\frac{dI_H}{dt} = ab_2 \left(\frac{I_M(t - \tau_H)}{N} \right) S_H(t - \tau_H) - \gamma I_H \quad (2)$$

$$\frac{dR_H}{dt} = \gamma I_H \quad (3)$$

$$\frac{dS_M}{dt} = b - \left(ab_1 \left(\frac{I_H}{N} \right) + \mu \right) S_M \quad (4)$$

$$\begin{aligned} \frac{dE_M}{dt} = & ab_1 \left(\frac{I_H}{N} \right) S_M - \mu E_M \\ & - ab_1 \left(\frac{I_H(t - \tau_M)}{N} \right) S_M(t - \tau_M) \mathcal{L}(\tau_M) \end{aligned} \quad (5)$$

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$$\begin{aligned} \frac{dI_M}{dt} = & ab_1 \left(\frac{I_H(t - \tau_M)}{N} \right) S_M(t - \tau_M) \ell(\tau_M) \\ & - \mu I_M \end{aligned} \quad (6)$$

The parameters are given in **Table A.1**, while the variables are given in **Table A.2**, which can be found in the Appendix.

However, the use of these equations proved to be unsuccessful in the appropriate modelling of the behavior of the disease, as can be seen in Figures 1A and 2A: while it would be expected for the number of infected humans to oscillate during the period of study, given that malaria is a recurring illness in the region, the model indicates that the number of infections would rapidly fall to 0, without occurrence of new cases.

In addition to that, it could also be seen that the mosquito population would also decline to near negligible levels, which could explain the lack of infections in the human population.

To overcome this effect, the SEI equations were modified in order for b_1 to only be used in the transference between compartments S and E , and a new parameter, b_3 to move mosquitos from E to I . Being inversely related to the incubation period, the definition of this rate can be found in **Table A.2**. In addition to that, the parameter $a(T)$ that was in use in the transition between exposed and infected also had to be removed, as no further bites occur at this point. The parameter l also had to be disassociated from the infection rate of exposed individual (b_3).

The SIR equations were also modified, to take into consideration population dynamics, including a r_H parameter as the net human increase rate per capita, which could be explained by births or immigration to the municipality. Given the positive growth of the studied population, a death rate was neglected for this study.

The updated equations used in the model are as following:

$$\frac{dS_H}{dt} = r_H N - ab_2 \left(\frac{I_M}{N} \right) S_H \quad (7)$$

$$\frac{dI_H}{dt} = ab_2 \left(\frac{I_M}{N} \right) S_H - \gamma I_H \quad (8)$$

$$\frac{dR_H}{dt} = \gamma I_H \quad (9)$$

$$\frac{dS_M}{dt} = b - \left(ab_1 \left(\frac{I_H}{N} \right) + \mu \right) S_M \quad (10)$$

$$\frac{dE_M}{dt} = ab_1 \left(\frac{I_H}{N} \right) S_M - (\mu + b_3 + l) E_M \quad (11)$$

$$\frac{dI_M}{dt} = b_3 E_M - \mu I_M \quad (12)$$

2.2. Base SEIRS-SEI model

Preliminary analysis of the baseline SIR-SEI framework revealed structural limitations in replicating endemic malaria transmission, as the system intrinsically tended toward a single outbreak followed by rapid fade-out. To rectify this and enhance biological fidelity, we extended the model to a SEIRS-SEI structure. This extension accounts for essential features of Plasmodium dynamics, such as the extrinsic incubation period in humans, represented by the Exposed compartment (E_H), and the decay of temporary acquired immunity, represented by the transition from Recovered to Susceptible ($R_H \rightarrow S_H$). These modifications are necessary to sustain the multi-year recurrence and seasonal oscillations characteristic of endemic regions.

The constructed model is as follows:

$$\frac{dS_H}{dt} = r_H N - ab_2 \left(\frac{I_M(t)}{N} \right) S_H + \omega R_H \quad (13)$$

$$\begin{aligned} \frac{dE_H}{dt} = & ab_2 \left(\frac{I_M(t)}{N} \right) S_H \\ & - ab_2 \left(\frac{I_M(t - \tau_H)}{N} \right) S_H(t - \tau_H) \end{aligned} \quad (14)$$

$$\begin{aligned} \frac{dI_H}{dt} = & ab_2 \left(\frac{I_M(t - \tau_H)}{N} \right) S_H(t - \tau_H) \\ & - \gamma I_H \end{aligned} \quad (15)$$

$$\frac{dR_H}{dt} = \gamma I_H - \omega R_H \quad (16)$$

$$\frac{dS_M}{dt} = \Lambda - \left(ab_1 \left(\frac{I_H(t)}{N} \right) + \mu \right) S_M \quad (17)$$

$$\begin{aligned} \frac{dE_M}{dt} = & ab_1 \left(\frac{I_H(t)}{N} \right) S_M - \mu E_M \\ & - ab_1 \left(\frac{I_H(t - \tau_M)}{N} \right) S_M(t - \tau_M) \ell \end{aligned} \quad (18)$$

$$\begin{aligned} \frac{dI_M}{dt} = & ab_1 \left(\frac{I_H(t - \tau_M)}{N} \right) S_M(t - \tau_M) \ell \\ & - \mu I_M \end{aligned} \quad (19)$$

Where Λ is the mosquito recruitment rate:

$$\Lambda = b(R, T) \cdot K \cdot \left(1 - \frac{S_M + E_M + I_M}{K} \right) \quad (20)$$

And K is the mosquito carrying capacity:

$$\begin{aligned} K = & M \times \text{Temp. factor} \times \text{Rain factor} \\ & \times \text{Seasonal factor} \end{aligned} \quad (21)$$

Where:

$$\text{Temp. factor} = \max \left(0.1, \min \left(1, \frac{T - 15}{15} \right) \right) \quad (22)$$

$$\text{Rain factor} = \min \left(1, \max \left(0, \frac{R}{30} \right) \right) \quad (23)$$

$$\text{Seasonal factor} = 2.5 + 2 \sin \left(2\pi \frac{t - 180}{365.25} \right) \quad (1 \leq t \leq 366) \quad (24)$$

3. Data description and processing

3.1. Epidemiological data

This study focuses on malaria cases caused by *Plasmodium vivax*, which accounts for the majority of malaria infections in Brazil and is the predominant species in the Amazon region [11, 12, 13].

Epidemiological data were obtained from the SIVEP-Malaria system [18], which provides daily records of notified malaria cases for all Brazilian municipalities from 2003 to 2023. The analysis was restricted to the period 2017–2023, for which complete epidemiological and environmental data were available.

3.2. Population estimates

The municipality of Manaus was selected due to its high incidence of *P. vivax* malaria, particularly in rural areas. However, detailed yearly estimates of rural population were not directly available from census data. To address this limitation, population estimates were derived using data from the Trajetorias Project [6], which provides incidence values aggregated over five-year periods.

Incidence is defined as

$$\text{Inc}(d, m, z, t_1, t_2) = \frac{\text{Cases}(d, m, z, t_1, t_2)}{\text{Pop}(m, z, (t_1 + t_2)/2) \cdot (t_2 - t_1)} \times 10^5, \quad (25)$$

where $\text{Cases}(d, m, z, t_1, t_2)$ denotes the number of reported cases for disease d in municipality m and zone z over the interval $[t_1, t_2]$, and $\text{Pop}(m, z, \cdot)$ represents the population at the midpoint of the interval.

This relationship was inverted to estimate rural population values from reported incidence and case counts. Using this approach, the rural population of Manaus was estimated to represent less than 1% of the total population, ranging between 0.3% and 0.6% of the total municipal population, while accounting for a disproportionately large share of malaria cases (approximately 30% to 40% over the study periods).

Yearly total population values for Manaus were obtained via interpolation based on official census data from the Brazilian Institute of Geography and Statistics (IBGE) for the years 2000, 2010, and 2022 [15, 16, 17]. Intermediate yearly values were constructed using linear interpolation, while values beyond 2022 were extrapolated based on observed trends.

3.3. Data consistency adjustments

During preprocessing, inconsistencies were identified in the Trajetorias dataset. In particular, incidence rates for the period 2015–2019 had been calculated using population estimates corresponding to 2004–2008, resulting in a temporal mismatch. These values were recalculated using population estimates consistent with the appropriate time interval.

Additionally, rural and urban population estimates for intermediate years had been derived using fixed proportions based on 2010 data. To better account for demographic changes, particularly urbanization, these estimates were revised. The rural population for 2006 was approximated as the mean between the reported values for 2000 and 2010, while the 2017 value was obtained through extrapolation based on the observed growth trend. These adjustments ensured temporal consistency in the population data used throughout the study.

4. Impacts of environmental factors

Environmental changes, particularly deforestation and forest fires, have been widely associated with the resurgence and spatial redistribution of malaria transmission in the Amazon region [20]. The removal of forest canopy alters local microclimatic conditions, increasing sunlight exposure and ambient temperature, which in turn favor the development of mosquito larvae and pupae. In particular, the primary malaria vector in the Amazon, *Anopheles darlingi*, is known to thrive in partially deforested environments, where sunlit water bodies promote larval growth and increase the availability of food sources such as algae [21, 22].

Beyond ecological effects, deforestation also influences human exposure to malaria. Activities such as logging, agriculture, and infrastructure development lead to increased human presence in recently altered environments, often in close proximity to mosquito breeding sites. This results in higher human–vector contact rates and facilitates transmission, especially in areas where both vector density and environmental suitability are elevated. Additionally, land-use changes such as agriculture may contribute to river sedimentation and the formation of artificial breeding sites, further enhancing mosquito proliferation.

Forest fragmentation has been shown to accelerate mosquito development cycles by increasing temperature and light availability, reducing the time from larval to adult stages and increasing vectorial capacity and survivorship [27, 28]. Similarly, forest fires, often used as a land-clearing mechanism, remove vegetation and modify habitat structure, contributing to long-term ecological changes that can influence mosquito populations and disease transmission dynamics [29, 30]. These processes may also increase the likelihood of subsequent fires by promoting understory drying and fuel accumulation [31].

In this context, deforestation and fire activity are incorporated into the model as external drivers that modify transmission dynamics. Rather than directly altering mosquito physiological parameters, these factors are interpreted as influencing human–vector contact rates and exposure, thereby affecting the effective transmission of the disease.

Deforestation data were obtained from the DETER system, which provides daily records of deforestation alerts by degradation type, reported in km² [34]. The data were aggregated to produce a complete daily time series of deforested area. Forest fire data were obtained from the Brazilian National Institute for Space Research (INPE), which provides daily records of detected fire hotspots [35]. Due to the absence of area-based measurements at this temporal resolution, the number of detected hotspots was used as a proxy for fire activity.

5. Mathematical analysis of the model

5.1. Basic Reproduction Number ($\mathcal{R}_0$)

To derive the $\mathcal{R}_0$ of the model, we define X_s as the set of all disease-free states,

$$X_s = \{x \geq 0 | x_i = 0, i = 1, \dots, m\},$$

where $X = (x_1, \dots, x_n)^T$, such that $x_i \geq 0$ represents the number of individuals in each compartment, and each function is assumed to be continuously differentiable at least twice in each variable (C^2) [36].

Rearranging the equations so that the first m contain the infected individuals, let $\mathcal{F}_i(x)$ be the rate of appearance of new infections in compartment i , $\mathcal{V}_i^+(x)$ be the rate of individuals entering compartment i by other means, and $\mathcal{V}_i^-(x)$ be the rate of individuals leaving compartment i . The disease transmission model consists of non-negative initial conditions together with the following system of equations:

$$\dot{x} = f_i(x) = \mathcal{F}_i(x) - \mathcal{V}_i(x), i = 1, \dots, n,$$

where $\mathcal{V}_i(x) = \mathcal{V}_i^-(x) - \mathcal{V}_i^+(x)$. Here, $F = \left[\frac{\partial \mathcal{F}_i(x_0)}{\partial x_j} \right]$ and $V = \left[\frac{\partial \mathcal{V}_i(x_0)}{\partial x_j} \right]$, where x_0 is a Disease-Free Equilibrium (DFE), and $1 \leq i, j \leq m$, and $\mathcal{R}_0 = \rho(FV^{-1})$.

The formulation for $\mathcal{R}_0$ in both model, as well as the complete SIR/SEI system is given by:

SIR: In this case, $m = 1$, and the compartments are arranged as $[I_H, S_H, R_H]$. Since $\mathcal{R}_0$ is calculated with normalized values, the necessary equations are multiplied by N to remove the denominator. Specifically, for the SIR case, as R_H is not used in any of the equations, it can be expressed solely in terms of S and I . Therefore:

- $\mathcal{F}_i(x)$: rate of entrance of infected individuals in compartment i

- $\mathcal{V}_i(x)^-$: rate of leaving compartment i
- $\mathcal{V}_i(x)^+$: rate of entering compartment i

So we have

$$\mathcal{F} = [ab_2 I_M S_H]$$

$$\mathcal{V}^- = [\gamma I_H]$$

$$\mathcal{V}^+ = [0]$$

$$\mathcal{V}_i(x) = \mathcal{V}_i(x)^- - \mathcal{V}_i(x)^+$$

Therefore,

$$\mathcal{V} = [I_H \gamma]$$

Hence

$$F = \frac{\partial \mathcal{F}}{\partial I_M} = [ab_2 S_H]$$

$$V = \frac{\partial \mathcal{V}}{\partial I_H} = [\gamma]$$

At the equilibrium, $[S_H^*, I_H^*] = [1, 0]$, so

$$F = [ab_2], V = [\gamma] \text{ and } \mathcal{R}_0 = \left| \frac{ab_2}{\gamma} \right|.$$

SEI: In this case, $m = 2$, and the compartments are arranged as $[E_M, I_M, S_M]$. Again, the necessary equations are multiplied by N to remove the denominator. Therefore:

$$\mathcal{F} = \begin{bmatrix} ab_1 I_H S_M \\ 0 \end{bmatrix}$$

$$\mathcal{V}^- = \begin{bmatrix} E_M(\mu + b_3 + l) \\ \mu I_M \end{bmatrix}$$

$$\mathcal{V}^+ = \begin{bmatrix} 0 \\ b_3 E_M \end{bmatrix}$$

$$\mathcal{V}_i(x) = \mathcal{V}_i(x)^- - \mathcal{V}_i(x)^+$$

Thus,

$$\mathcal{V} = \begin{bmatrix} E_M(\mu + b_3 + l) \\ \mu I_M - b_3 E_M \end{bmatrix}$$

Hence

$$\begin{aligned} F &= \frac{\partial \mathcal{F}}{\partial E_M, I_H} \\ &= \begin{bmatrix} \frac{\partial ab_1 I_H S_M}{\partial E_M} & \frac{\partial ab_1 I_H S_M}{\partial I_H} \\ \frac{\partial 0}{\partial E_M} & \frac{\partial 0}{\partial I_H} \end{bmatrix} \\ &= \begin{bmatrix} 0 & ab_1 S_M \\ 0 & 0 \end{bmatrix} \end{aligned}$$

$$V = \frac{\partial \mathcal{V}}{\partial E_M, I_M}$$

$$= \begin{bmatrix} \frac{\partial E_M(\mu + b_3 + l)}{\partial \mu I_M - b_3 E_M} & \frac{\partial E_M(\mu + b_3 + l)}{\partial I_M} \\ \frac{\partial E_M}{\partial \mu I_M - b_3 E_M} & \frac{\partial E_M}{\partial I_M} \end{bmatrix}$$

$$= \begin{bmatrix} \mu + b_3 + l & 0 \\ -b_3 & \mu \end{bmatrix}$$

At the equilibrium, $[S_M^*, E_M^*, I_M^*] = [1, 0, 0]$, so

$$F = \begin{bmatrix} 0 & ab_1 \\ 0 & 0 \end{bmatrix},$$

$$V = \begin{bmatrix} \mu + b_3 + l & 0 \\ -b_3 & \mu \end{bmatrix}$$

and

$$\mathcal{R}_0 = \left| \frac{ab_1 b_3}{(b_3 + l + \mu)\mu} \right|.$$

SIR/SEI: In this case, $m = 3$, and the compartments are arranged as $[I_H, E_M, I_M, S_H, S_M]$. Once again, the necessary equations are multiplied by N to remove the denominator. Therefore:

$$\mathcal{F} = \begin{bmatrix} ab_2 I_M S_H \\ ab_1 I_H S_M \\ 0 \end{bmatrix}$$

$$\mathcal{V}^- = \begin{bmatrix} \gamma I_H \\ E_M(\mu + b_3 + l) \\ \mu I_M \end{bmatrix}$$

$$\mathcal{V}^+ = \begin{bmatrix} 0 \\ 0 \\ b_3 E_M \end{bmatrix}$$

$$\mathcal{V}_i(x) = \mathcal{V}_i(x)^- - \mathcal{V}_i(x)^+$$

Thus,

$$\mathcal{V} = \begin{bmatrix} I_H \gamma \\ E_M(\mu + b_3 + l) \\ \mu I_M - b_3 E_M \end{bmatrix}$$

Hence

$$F = \frac{\partial \mathcal{F}}{\partial I_H, E_M, I_M} = \begin{bmatrix} 0 & 0 & ab_2 S_H \\ ab_1 S_M & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$V = \frac{\partial \mathcal{V}}{\partial I_H, E_M, I_M} = \begin{bmatrix} \gamma & 0 & 0 \\ 0 & b_3 + l + \mu & 0 \\ 0 & -b_3 & \mu \end{bmatrix}$$

At the equilibrium,

$$[S_H^*, S_M^*, I_H^*, E_M^*, I_M^*] = [1, 1, 0, 0, 0],$$

so

$$F = \begin{bmatrix} 0 & 0 & ab_2 \\ ab_1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

$$V = \begin{bmatrix} \gamma & 0 & 0 \\ 0 & b_3 + l + \mu & 0 \\ 0 & -b_3 & \mu \end{bmatrix}$$

and

$$\mathcal{R}_0 = \left| \sqrt{\frac{a^2 b_1 b_2 b_3}{(b_3 + l + \mu)\gamma\mu}} \right| = \sqrt{\mathcal{R}_{0SIR} \times \mathcal{R}_{0SEI}}$$

5.2. Understanding relations

Having the necessary epidemiological and environmental datasets ready for use, the next step taken was to verify the correlation between the number of cases and forest areas impacted by deforestation and fires, in addition to the correlation with temperature and precipitation as well.

Using the Pandas library on a Python environment, the correlation between monthly number of cases caused by *P. vivax* in Manaus and deforested area was found to be approximately -0.11, and 0.137 between cases and burnt area. In relation to climatic data, the correlation with temperature was found to be around 0.404, and with precipitation to be -0.365. Both correlation tables were taken in relation to monthly data.

On the other hand, taking the correlation between the incidence of cases in relation to the seasons, deforested and burnt areas, that was found to be 0.722, 0.314 and 0.818, respectively. Looking for the correlation in relation to seasons, temperature and precipitation, that was found to be 0.687, 0.839 and -0.883, respectively.

6. Results

The initial implementation of the model using the original functional forms and parameter values reported in [7] did not reproduce biologically consistent transmission dynamics, particularly in the mosquito population. As illustrated in Figures 1–3A, the human SIR component exhibits a monotonic decline in the number of infected individuals, suggesting that the infection fails to persist in the population under the given parametrization. More critically, the corresponding SEI mosquito model yields nonphysical behavior,

with the susceptible mosquito population attaining negative values and exhibiting oscillatory dynamics.

These inconsistencies arise from the functional forms adopted for key entomological parameters, which do not explicitly enforce biologically meaningful bounds. In particular, probabilities and rates are not constrained to lie within the interval $[0, 1]$, leading to violations of positivity and instability in the numerical solutions.

To address this issue, the functional forms were modified to ensure that all probabilities and rates remain within their admissible ranges. The resulting dynamics, shown in Figure 3A, exhibit qualitatively different behavior, with the mosquito population rapidly declining toward extinction. While this modification restores mathematical consistency, it also highlights a mismatch between the model structure and the expected persistence of malaria in endemic settings, motivating the need for further refinement of the transmission and demographic assumptions.

Before including the environmental factors on the used model (7-12), its results were compared to a model based on SIVEP data, keeping use of equations (7, 10-12), but using

$$I_H(t) = \sum_{t=\max\left(0, k-\frac{1}{\gamma}+1\right)}^{0 \leq k \leq 1825} \# \text{ of cases} \quad (26)$$

$$R_H(t) = \max\left(0, N(1 + r_H t) - S_H(t) - I_H(t)\right) \quad (27)$$

$I_H(t)$ now being the moving sum of cases over a period of 120 days, as it is the average duration of infectiousness, and $R_H(t)$ being the difference between the initial population and the susceptible and infected population at each day, with the addition of the daily births. The result can be seen in the graph below.

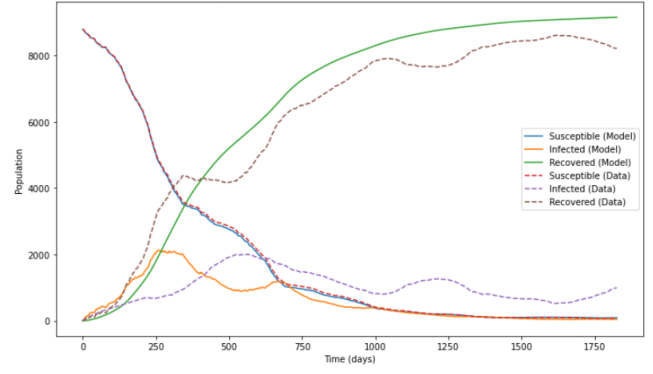


Figure 1: Comparison between observed data of infections and model estimation of I_H .

It is possible to see that the model gives a close approximation to the format of the observed number of infections and recoveries, although the infections appear to increase faster and die out given the model, a behavior that is not seen on the actual data.

At this point, a linear combination was implemented as a replacement to the use of simply T and R , in order to take into consideration the effects of deforestation and fires, with the α -function $\alpha_1 D + \alpha_2 F + \alpha_3 T + \alpha_4 R$, with $\alpha_0 = [0, 0, 1, 1]$ representing the primary version of the model.

7. Computational Implementation

All model simulations were performed using Python version 3.12.7 (conda-forge distribution). Data manipulation and analysis were performed using Pandas version 3.0.2, while numerical computations and array operations were carried out with NumPy version 2.4.2.

The systems of time-delay ordinary differential equations (1-6) were solved numerically using the explicit Runge-Kutta method of order 8 as implemented in the `solve_ivp` function from SciPy version 1.17.1. All computations were performed on a laptop running **Windows 11** (64-bit) equipped with a **12th Gen Intel Core i5-1235U** processor (10 cores, 12 threads, 1.30 GHz base frequency) and **16 GB RAM**.

The source code and data required to reproduce all figures are available at https://github.com/RaphaLevy/Study_on_malaria_dynamics_v2.

References

- [1] P. F. P. Pimenta et al., An overview of malaria transmission from the perspective of Amazon *Anopheles* vectors, *Memórias do Instituto Oswaldo Cruz*, Vol. 110 (1) (2015), 23-47, <https://doi.org/10.1590/0074-02760140266>.
- [2] G. A. Josling, K. C. Williamson, M. Llinás, Regulation of Sexual Commitment and Gametocytogenesis in Malaria Parasites, *Annual Review of Microbiology*, Vol. 72 (1) (2018), 501-519, <https://doi.org/10.1146/annurev-micro-090817-062712>.
- [3] World Health Organization / World Malaria Report 2022, Global Malaria Programme, accessed on 29 August 2024, <https://www.who.int/publications/i/item/9789240064898>.
- [4] M. da Silva-Nunes et al., Malaria on the Amazonian frontier: transmission dynamics, risk factors, spatial distribution, and prospects for control, *American Journal of Tropical Medicine and Hygiene*, Vol. 79 (4) (2008), 624-635, <https://doi.org/10.4269/ajtmh.2008.79.624>.
- [5] M. da Silva-Nunes et al., Amazonian malaria: Asymptomatic human reservoirs, diagnostic challenges, environmentally driven changes in mosquito vector populations, and the mandate for sustainable control strategies, *Acta Tropica*, Vol. 121 (3) (2012), 281-291, <https://doi.org/10.1016/j.actatropica.2011.10.001>.
- [6] A. C. Rorato, A. P. Dal'Asta, R. M. Lana et al., Trajetórias: a dataset of environmental, epidemiological, and economic indicators for the Brazilian Amazon, *Scientific Data*, Vol. 10 (1) (2023), 65, <https://doi.org/10.1038/s41597-023-01962-1>.
- [7] P. E. Parham, E. Michael, Modelling Climate Change and Malaria Transmission, *Modelling Parasite Transmission and Control*, *Advances in Experimental Medicine and Biology*, Vol. 673 (2010), 184-199, https://doi.org/10.1007/978-1-4419-6064-1_13.
- [8] N. J. Arisco, C. Peterka, C. Diniz, B. H. Singer, M. C. Castro, Ecological change increases malaria risk in the Brazilian Amazon, *Proc. Natl. Acad. Sci.*, Vol. 121 (44) (2024) <https://doi.org/10.1073/pnas.2409583121>.
- [9] G. C. McCord, Malaria ecology and climate change, *The European Physical Journal Special Topics*, Vol. 225 (3) (2016), 459-470, <https://doi.org/10.1140/epjst/e2015-50097-1>.
- [10] S. W. Lindsay, M. H. Birley, Climate change and malaria transmission, *Annals of Tropical Medicine & Parasitology*, Vol. 90 (1996), 573-588, <https://doi.org/10.1080/00034983.1996.11813087>.
- [11] J. Oliveira-Ferreira, M. V. Lacerda, P. Brasil et al., Malaria in Brazil: an overview., *Malaria Journal*, Vol. 9 (1) (2010), <https://doi.org/10.1186/1475-2875-9-115>.
- [12] C. T. Codeço, A. P. Dal'Asta, A. C. Rorato, R. M. Lana, T. C. Neves, C. S. Andreazzi, M. Barbosa, M. I. S. Escada, D. A. Fernandes, D. L. Rodrigues, I. C. Reis, M. da Silva-Nunes, A. B. Gontijo, F. C. Coelho, A. M. V. Monteiro, Epidemiology, Biodiversity, and Technological Trajectories in the Brazilian Amazon: From Malaria to COVID-19, *Frontiers in Public Health*, Vol. 9 (2021), <https://doi.org/10.3389/fpubh.2021.647754>.
- [13] A. K. R. Galardo, M. Arruda, et al., Malaria vector incrimination in three rural riverine villages in the Brazilian Amazon, *The American Journal of Tropical Medicine and Hygiene*, Vol. 76 (3) (2007), 461-469, <https://www.ajtmh.org/view/journals/tpmd/76/3/article-p461.xml>.
- [14] Tabnet, Ministério da Saúde, accessed on 12 August 2024, <http://tabnet.datasus.gov.br/cgi/tabcgi.exe?ibge/cnv/popam.def>.
- [15] Biblioteca IBGE, 2000, accessed on 11 November 2024, <https://biblioteca.ibge.gov.br/index.php/biblioteca-catalogo?view=detalhes&id=7308>.
- [16] IBGE, Censo Demográfico, 2010, accessed on 11 November 2024, <https://www.ibge.gov.br/estatisticas/sociais/populacao/9662-censo-demografico-2010.html?=&t=resultados>.
- [17] IBGE, Brasil / Amazonas / Manaus, 2022, accessed on 11 November 2024, <https://cidades.ibge.gov.br/brasil/am/manaus/panorama>.
- [18] SIVEP Malaria, accessed on 10 October 2024, <https://files.deeplearn.ltd/a/hnjvpj>.
- [19] Mosqlimate Data API / Datastore / climate, accessed on 12 August 2024, <https://api.mosqlimate.org/datastore/>.
- [20] D. E. Norris, Mosquito-borne Diseases as a Consequence of Land Use Change, *EcoHealth*, Vol. 1 (1) (2004), 19-24, <https://doi.org/10.1007/s10393-004-0008-7>.
- [21] Anopheles, Fiocruz Rondônia, accessed on 15 August 2024, <https://www.rondonia.fiocruz.br/pivem/anopheline/>.
- [22] M. da Silva-Nunes, Environmental changes impact in malaria transmission and prospects for the disease control in Brazilian Amazon rural settlements, *Oecologia Australis*, Vol. 14 (3) (2010), 603-622, <https://revistas.ufjf.br/index.php/oa/article/view/7101>.
- [23] J. D. Charlwood, Biological variation in *Anopheles darlingi* Root, *Memórias do Instituto Oswaldo Cruz*, Vol. 91 (4) (1996), 391-398, <https://doi.org/10.1590/S0074-02761996000400001>.
- [24] G. R. de Lima, A. N. Arcos, E.V. dos Santos, R. C. Simões, C.A.P. Lima, W. P. Tadei, Anopheles Larval Abundance in Artificial Breeding Sites in the East Zone of Manaus, Amazon, *South American Journal of Basic Education, Technical and Technological*, Vol. 8 (1) (2021), 35-47, <https://periodicos.ufac.br/index.php/SAJEBTT/article/view/3399>.
- [25] C. M. Souza, Jr., J. Z. Shimbo, M. R. Rosa, L. L. Parente, A. A. Alencar, B. F. T. Rudorff, H. Hasenack, M. Matsumoto, L. G. Ferreira, P. W. M. Souza-Filho, et al, Reconstructing Three Decades of Land Use and Land Cover Changes in Brazilian Biomes with Landsat Archive and Earth Engine, *Remote Sensing*, Vol.12 (17) (2020), <https://doi.org/10.3390/rs12172735>.
- [26] A. J. MacDonald, E. A. Mordecai, Amazon deforestation drives malaria transmission, and malaria burden reduces forest clearing, *Proceedings of the National Academy of Sciences*, Vol. 116 (44) (2019), <https://doi.org/10.1073/pnas.1905315116>.
- [27] M. da Silva-Nunes, Environmental changes impact in malaria transmission and prospects for the disease control in Brazilian Amazon rural settlements, *Oecologia Australis*, Vol. 14 (3) (2010), 603-622, <https://doi.org/10.4257/oeco.2010.1403.02>.
- [28] L. S. M. Chaves, E. S. Bergo, J. E. Conn, G. Z. Laporta, P. R. Prist, M. A. M. Sallum, Anthropogenic landscape decreases mosquito biodiversity and drives malaria vector proliferation in the Amazon rainforest, *PLOS ONE*, Vol. 16 (1) (2021), <https://doi.org/10.1371/journal.pone.0245087>.
- [29] K. R. Kirby, W. F. Laurance, A. K. L. M. Albernaz, G. Schroth, P. M. Fearnside, S. Bergen, E. M. Venticinque, C. Da Costa, The future of

deforestation in the Brazilian Amazon, *Futures*, Vol. 38 (4), 432–453 (2006), <https://doi.org/10.1073/pnas.1905315116>.

- [30] C. F. Lima, F. T. P. Torres, L. J. Minette, F. A. Lima, R. C. A. Lima et al, Is there a relationship between forest fires and deforestation in the Brazilian Amazon?, *PLOS ONE*, Vol. 19 (6), <https://doi.org/10.1371/journal.pone.0306238>.
- [31] P. M. Fearnside, Conservation Policy in Brazilian Amazonia: Understanding the Dilemmas, *World Development*, Vol. 31 (5), 757-779 (2003), [https://doi.org/10.1016/S0305-750X\(03\)00011-1](https://doi.org/10.1016/S0305-750X(03)00011-1).
- [32] L. Ziccardi, A. Yanai, P. Fearnside, S. Stark, Deforestation in the Brazilian Amazon: Threats and opportunities, Research Square, PREPRINT (Version 1) (2024), <https://doi.org/10.21203/rs.3.rs-4591425/v1>.
- [33] G1, M. Castro, "Amazonas registra 21,6 mil queimadas em 2024 e tem o pior índice em 26 anos, aponta Inpe", accessed on 28 September 2024, <https://g1.globo.com/am/amazonas/noticia/2024/09/24/amazonas-registra-216-mil-queimadas-em-2024-e-tem-o-pior-indice-em-26-anos-aponta-inpe.gh.html>.
- [34] TerraBrasilis, DETER (Notices), Analyses - Amazônia Biome, Daily notices of DETER project area, accessed on 17 March 2026, <https://terrabrasilis.dpi.inpe.br/app/dashboard/alerts/biomes/amazonia-nb/daily/>.
- [35] Data_Basis.org, Fire Database, National Institute for Space Research (INPE), accessed on 17 March 2026, <https://data-basis.org/dataset/f06f3cdc-b539-409b-b311-1ff8878fb8d9?table=a3696dc2-4dd1-4f7e-9769-6aa16a1556b8>.
- [36] P. van den Driessche, J. Watmough, Reproduction numbers and sub-threshold endemic equilibria for compartmental models of disease transmission, *Mathematical Biosciences*, Vol. 180 (2002), 29-48, [https://doi.org/10.1016/S0025-5564\(02\)00108-6](https://doi.org/10.1016/S0025-5564(02)00108-6).

A. Appendix

A.1. Tables

Table A.1

List of model parameters.

Parameter	Definition	Formulation
$b(R, T)$	Mosquito birth rate (days ⁻¹)	$B_E p_E(R) p_L(R, T) p_P(R) / (\tau_E + \tau_L(T) + \tau_P)$
$a(T)$	Biting rate (days ⁻¹)	$(T - T') / D_1$
$\mu(T)$	Mosquito mortality rate per capita (days ⁻¹)	$-\log(p(T))$
$\tau_M(T)$	Duration of sporozoite cycle (days)	$DD / (T - T_{min})$
$\tau_L(T)$	Duration of larval development phase (days)	$1 / (c_1 T + c_2)$
$p(T)$	Daily mosquito survival rate	$e^{(-1 / (AT^2 + BT + C))}$
$p_L(R)$	Probability of larval survival dependent on rainfall	$(4p_{ML} / R_L^2) R(R_L - R)$
$p_L(T)$	Probability of larval survival dependent on temperature	$e^{-(c_1 T + c_2)}$
$p_L(R, T)$	Probability of larval survival	$p_L(R) p_L(T)$
$l(\tau_M)(T)$	Probability of mosquito survival during sporozoite cycle (days ⁻¹)	$p(T)^{\tau_M(T)}$
$M(t)$	Total population of mosquitos	$S_M(t) + E_M(t) + I_M(t)$
$N(t)$	Total population of humans	$S_H(t) + I_H(t) + R_H(t)$

Table A.2

List of model variables.

Symbol	Definition	Units
b_1	Proportion of bites from susceptible mosquitoes on infected humans that result in infection	Dimensionless
b_2	Proportion of bites from infected mosquitoes on susceptible humans that result in infection	Dimensionless
γ	1/Average duration of infectiousness in humans	days ⁻¹
T_1	Mean temperature in the absence of seasonality	°C
T_2	Amplitude of seasonal variability in temperature	Dimensionless
ω_1	Angular frequency of seasonal oscillations in temperature	months ⁻¹
ϕ_1	Phase lag of temperature variability (phase shift)	Dimensionless
R_1	Average monthly precipitation in the absence of seasonality	mm
R_2	Amplitude of seasonal variability in precipitation	Dimensionless
ω_2	Angular frequency of seasonal oscillations in precipitation	months ⁻¹
ϕ_2	Phase lag of precipitation variability (phase shift)	Dimensionless
B_E	Number of eggs laid per adult per oviposition	Dimensionless
p_{ME}	Maximum probability of egg survival	Dimensionless
p_{ML}	Maximum probability of larval survival	Dimensionless
p_{MP}	Maximum probability of pupal survival	Dimensionless
τ_E	Duration of the egg development phase	days
τ_P	Duration of the pupal development phase	days
τ_H	Latent period of infection within humans	days
R_L	Rainfall threshold until breeding sites are eliminated, removing immature individuals: 32.26 [8]	mm
T_{min}	Minimum temperature, below which there is no development of the parasite: 14.5	°C
DD	Degree-days for parasite development. "Sum of heat" for maturation: 105 [9, 10]	°C days
A	Empirical sensitivity parameter	(°C ² days) ⁻¹
B	Empirical sensitivity parameter	(°C days) ⁻¹
C	Empirical sensitivity parameter	days ⁻¹
D_1	Temperature scaling factor of the biting rate	°C days
c_1	Empirical sensitivity parameter	(°C days) ⁻¹
c_2	Empirical sensitivity parameter	days ⁻¹
T'	Biting rate temperature threshold ¹	°C
b_3	Infection rate in exposed mosquitoes ($1/\tau_M(T)$)	days ⁻¹
r_H	Net daily recruitment rate per capita	days ⁻¹

¹ In Parham and Michael [7], this parameter is denoted T_1 . We rename it here to avoid confusion with the mean temperature in the absence of seasonality, also denoted T_1 in the original paper.

A.2. Figures

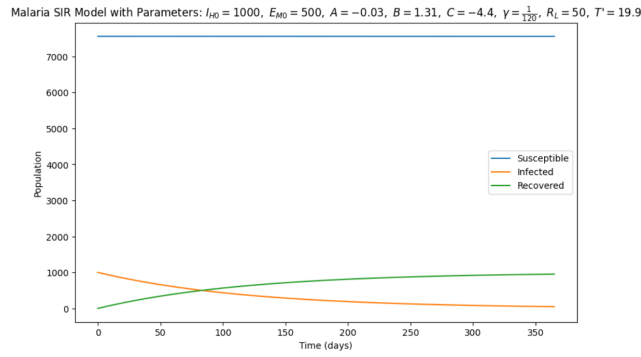


Figure 1A: Simulation of the baseline SIR model using the original parameterization from [7]. The dynamics show a monotonic decline in the number of infected individuals, indicating the absence of sustained transmission.

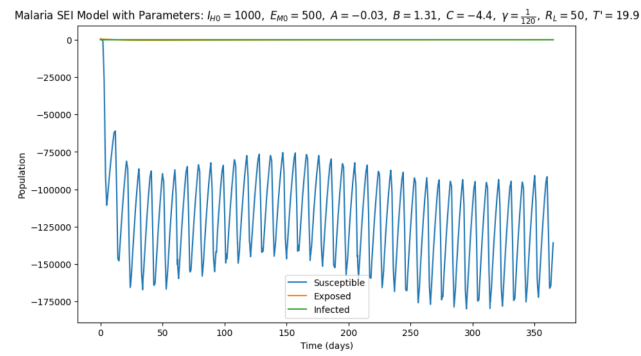


Figure 2A: Simulation of the baseline SEI mosquito model using the original parameterization from [7]. The susceptible mosquito population exhibits oscillatory behavior and attains negative values, indicating a loss of biological and mathematical consistency.

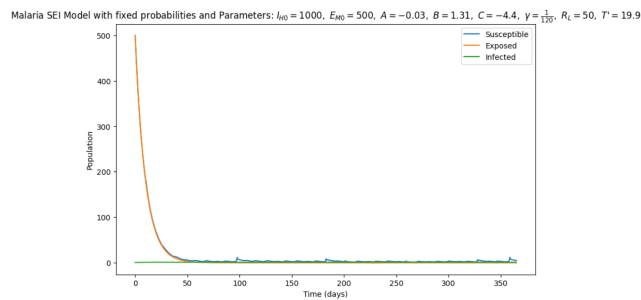


Figure 3A: Simulation of the SEI mosquito model after modifying parameter functions to enforce biologically admissible bounds on probabilities and rates. Although positivity is restored, the mosquito population rapidly declines toward extinction, suggesting that additional mechanisms are required to sustain endemic transmission.